1.06 Exponentials and Logarithms				
1.06a	Properties of the exponential function	a) Know and use the function a^x and its graph, where a is positive. Know and use the function e^x and its graph. *Examples may include the comparison of two population models or models in a biological or financial context. The link with geometric sequences may also be made.*		MF1
1.06b	Gradient of e^{kx}	b) Know that the gradient of e^{kx} is equal to ke^{kx} and hence understand why the exponential model is suitable in many applications. *See 1.07j for explicit differentiation of* e^x.		MF2

OCR Ref.	Subject Content	Stage 1 learners should ...	Stage 2 learners additionally should ...	DfE Ref.
1.06c 1.06d 1.06e	Properties of the logarithm	c) Know and use the definition of $\log_a x$ (for $x > 0$) as the inverse of a^x (for all x),where a is positive. *Learners should be able to convert from index to logarithmic form and vice versa as $a = b^c \Leftrightarrow c = \log_b a$.* *The values* $\log_a a = 1$ *and* $\log_a 1 = 0$ *should be known.* d) Know and use the function $\ln x$ and its graph. e) Know and use $\ln x$ as the inverse function of e^x. *e.g. In solving equations involving logarithms or exponentials.* *The values* $\ln \mathrm{e} = 1$ *and* $\ln 1 = 0$ *should be known.*		MF3
1.06f	Laws of logarithms	f) Understand and be able to use the laws of logarithms: 1. $\log_a x + \log_a y = \log_a (xy)$ 2. $\log_a x - \log_a y = \log_a \left(\frac{x}{y}\right)$ 3. $k \log_a x = \log_a x^k$ (including, for example, $k = -1$ and $k = -\frac{1}{2}$). *Learners should be able to use these laws in solving equations and simplifying expressions involving logarithms.* [*Change of base is excluded*.]		MF4
1.06g	Equations involving exponentials	g) Be able to solve equations of the form $a^x = b$ for $a > 0$ *Includes solving equations which can be reduced to this form such as $2^x = 3^{2x-1}$, either by reduction to the form $a^x = b$ or by taking logarithms of both sides.*		MF5

OCR Ref.	Subject Content	Stage 1 learners should ...	Stage 2 learners additionally should ...	DfE Ref.
1.06h	Reduction to linear form	h) Be able to use logarithmic graphs to estimate parameters in relationships of the form $y = ax^n$ and $y = kb^x$, given data for x and y. *Learners should be able to reduce equations of these forms to a linear form and hence estimate values of a and n, or k and b by drawing graphs using given experimental data and using appropriate calculator functions.*		MF6
1.06i	Modelling using exponential functions	i) Understand and be able to use exponential growth and decay and use the exponential function in modelling. *Examples may include the use of* e *in continuous compound interest, radioactive decay, drug concentration decay and exponential growth as a model for population growth. Includes consideration of limitations and refinements of exponential models.*		MF7

OCR Ref.	Subject Content	Stage 1 learners should ...	Stage 2 learners additionally should ...	DfE Ref.